Neuronal Tumors, Medulloblastoma, and Meningioma

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Define and describe the pathology and clinical presentation of meningioma.
2. Define and describe the pathology and clinical presentation of medulloblastoma.
3. Define neuronal tumors, including ganglioglioma, and describe the pathology and clinical presentation.

CASE CONNECTION

LW is a 62-year-old female who presents to the emergency department after a motor vehicle crash and head trauma. She is alert and oriented, describing pain where her head hit the windshield. She has no other discomfort. You note no abnormal findings on physical examination. You order a CT scan of the head as part of the trauma protocol; a 2.5-cm dural-based parasagittal mass is seen.

What is the likely diagnosis? What is the prognosis? Consider your answers as you read, and we'll revisit LW at the end of the brick.

GO TO CONCLUSION

What Are Neuronal Tumors, Medulloblastoma, and Meningioma?

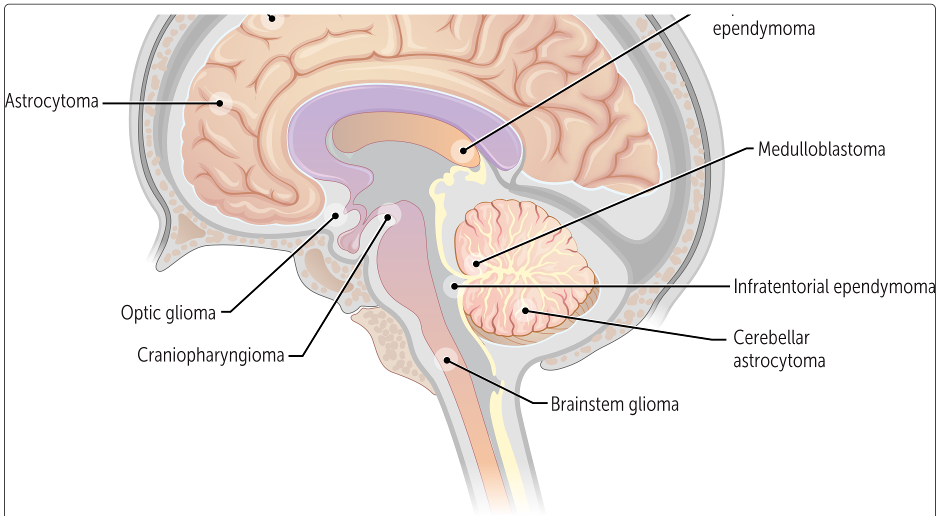
TRUCTOR NOTE

Please note that in the part 1 and part 2 discussions of all brain tumors, benign or malignant, the assumption is that they are primary. However, some sources indicate that almost half of brain tumors in hospitalized patients are metastatic tumors from other organs, such as the lung or skin (melanoma). Please see the required supplemental handout for this topic posted on Canvas.

MARIA PLUMMER

Neuronal tumors, medulloblastoma, and meningioma are considered primary brain tumors that originate within the brain. Although often malignant, primary brain tumors seldom spread to distant organs. In fact, more brain tumors are metastases from other organs than primaries. Without much proliferation, there is less potential for mutagenesis and thus carcinogenesis. As brain tumors grow, they eventually compress or invade brain tissue, becoming symptomatic or even life-threatening.

These three types of primary brain tumors originate from different cell lineages: meningiomas from the meninges, medulloblastomas from embryonic tissue, and neuronal tumors from neural elements (Figure 1).



QUIZ

Tap image for quiz

Figure 1

What Are Meningiomas?

Meningiomas account for 20%-30% of primary brain tumors, making them the most common of all primary brain tumors. They are just what they sound like—tumors of the meningeal tissue (dura mater, arachnoid, and pia mater). Meningiomas arise from the arachnoid cells. About 20,000 new meningiomas are diagnosed yearly, with a 2:1 female-to-male predominance. They are typically slow-growing, peripheral masses mainly in adults. They have a good prognosis with surgery because they are not attached to the brain. If complete resection is not possible, or if surgery is not even an option, radiation therapy is the second-line treatment.

TRUCTOR NOTE

Infiltrating astrocytomas and glioblastomas make up about 80% of primary brain tumors, (Robbins). As stated in the Scholar Rx bricks for Gliomas, glioblastomas make up about half of primary adult brain tumors. The statement above, indicating that meningiomas account for the most common type of primary brain tumor, does not appear accurate. Perhaps "primary benign brain tumor" may be more accurate Remember, sources do vary when describing percentages.

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Characteristics

These tumors grow so slowly, and they can often become quite large before producing symptoms, because the brain has plenty of time to adjust to the added pressure. Eventual symptoms include headache, seizure, vision problems, and focal neurological deficits (eg, weakness in an extremity). Symptoms are largely dependent on the location of the tumor. The most common locations include parasagittal, falcine, and convexities (Figure 2).

TRUCTOR NOTE

Please note that because of the presence of progesterone receptors on meningiomas, they may increase in size during pregnancy.

Menigiomas are often described as "dural-based".

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QUIZ

Tap image for quiz

Figure 2

Neurofibromatosis type 2 is a predisposing factor for meningioma, and molecular testing has revealed up to 80% of these tumors may have mutations in the *NF2* gene.

TRUCTOR NOTE

please see handout

MARIA PLUMMER

A very common imaging finding for meningioma is a “dural tail,” as they grow from the meninges, and indicates their extra-axial origin. Grossly, these tumors are well circumscribed.

Histology

On histology, you can see spindle cells arranged in a whorled pattern and lamellated calcifications known as psammoma bodies (Figure 3).

STRUCTOR NOTE

> This is only one histological subtype of meningioma but it is probably the most common - there are several others but just be familiar with this one.
>
> **MARIA PLUMMER**

QUIZ

Tap image for quiz

Figure 3

Psammoma bodies occur in a few other important conditions that should be remembered and differentiated from one another.

Conditions with psammoma bodies on histology = **PS**a**MM**oma:

Papillary carcinoma of the thyroid

Serous papillary cystadenocarcinoma of the ovary

Meningiomas

Malignant mesothelioma

What are the histologic features of meningiomas?

What Are Medulloblastomas?

Medulloblastomas are the most common malignant brain tumor in children. An embryonal tumor is embryonic tissue remaining in the brain after birth that becomes neoplastic. Symptoms develop over weeks to months, with the definitive diagnosis usually confirmed within 3 months of symptom onset. Medulloblastomas primarily occur in the cerebellum. Typical symptoms include headache, cranial nerve VI palsies, and cerebellar ataxia. Because these tumors can grow into the fourth ventricle, hydrocephalus is also a possibility.

Characteristics

Medulloblastomas are high-grade tumors. Usually treatment consists of resection followed by radiation and chemotherapy. The prognosis largely depends on resection and molecular characteristics. Metastases are common, warranting whole neuraxis imaging. Medulloblastomas are the most common embryonal tumors in children, with 70% being diagnosed in children younger than 10 years. Children with average risk medulloblastoma have a 5-year disease-free survival rate of 70% to 80%. Overall, children younger than 3 years have poor outcomes.

STRUCTOR NOTE

The molecular characteristics which make up the sub-types of medulloblastomas are important in prognosis

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Histology

On histology, there are small, round cells from an undifferentiated embryonal lineage. The undifferentiated characteristics of this tumor make sense: medulloblastomas are derived from embryonic tissue. Furthermore, cells often surround a neuropil (network of nerve fibers, synapses, and glial cells) termed Homer-Wright rosettes, which are characteristic of all tumors with neuroblastic proliferation (Figure 4).

QUIZ

Tap image for quiz

Figure 4

How do medulloblastomas appear on histology?

What Are Neuronal Tumors?

Neuronal tumors comprise abnormal glial and neuronal elements. They can be classified as pure neuronal cell tumors (eg, gangliocytomas, which are exceedingly rare and not discussed here) or mixed neuronal cell tumors (eg, ganglioglioma). Overall, neuronal tumors make up about 1% of all brain neoplasms.

Most neuronal tumors are located within the brain parenchyma, producing characteristic deficits based on location. As neuronal tumors grow, they disrupt the normal brain architecture and thus the neuronal networks. As a result, patients often present with seizures. Neuronal cell tumors are usually low grade and slow growing, so they have a favorable prognosis. They are usually cured with surgery alone.

As the name suggests, **gangliogliomas** are mixed neuronal cell tumors that contain a disorganized arrangement of ganglion cells and glia (indicating neuronal support cells). These tumors are often slow growing and therefore usually considered benign (World Health Organization [WHO] grade I).

Gangliogliomas have an incidence of 4% of all primary brain tumors in children and 1% of all primary brain tumors in adults. Up to 10% can become malignant. They tend to occur in the temporal lobes and present as new-onset seizures in children and young adults. Treatment of these tumors is resection. The prognosis is good, with a 10-year survival rate of 84%-93%.

Neoplastic ganglion cells with a glial component are seen on histology. The glial component of the tumor accounts for most of the tumor proliferation. The neuronal component exhibits binucleated and dysmorphic neurons with peripheral Nissl bodies. Nissl bodies are simply the rough endoplasmic reticulum of the neuron. These neoplastic neurons tend to cluster. If the glial component is astrocytic in nature, Rosenthal fibers might be present. Calcifications are another common feature seen on histology.

CASE CONNECTION BACK TO INTRODUCTION

Thinking back to LW, what is the likely diagnosis? What is the prognosis?

According to the CT appearance, you diagnose an asymptomatic meningioma, a benign brain tumor of the meninges and the most common primary brain tumor. As in LW's case, meningiomas can present asymptomatically and can be discovered on imaging studies obtained for other reasons. Prognosis is excellent when complete surgical resection is possible. LW says, "When I heard the words 'brain tumor,' I freaked out, but now I feel better. I appreciate the time you spent describing this to me. I'm okay with the plan for surgery."

Summary

- Meningiomas are peripherally located tumors composed of meningeal tissue that often appear as spindle cells arranged in a whorled pattern with dispersed psammoma bodies.
- Medulloblastomas are the most common embryonal tumor in children, typically found in the cerebellum; they comprise small, round cells forming Homer-Wright rosettes and have variable prognosis (good with treatment, dismal without).
- Neuronal tumors can be classified as pure neuronal cell tumors or mixed neuronal cell tumors.
- Gangliogliomas are mixed neuronal cell tumors composed of neoplastic ganglion and glial cells; accordingly, gangliogliomas have histologic characteristics resembling both gangliocytomas (eg, dysplastic and often binucleated with Nissl bodies) and gliomas (eg, Rosenthal fibers).

Review Questions

1. **A 3-year-old boy is seen for progressive headaches. His mom says that he started complaining of headaches a month ago. Furthermore, the mom says her son has recently started walking "funny." On neurologic exam, the boy misjudges the location of the clinician's finger when moving his finger from his nose and the clinician's finger with his left hand, a sign known as dysmetria. This child's symptoms are most concerning for a tumor in which location?**

   A. Cerebellum

   B. Frontal lobe

   C. Parietal lobe